Eduardo de Jesús Dávila Meza, Ph.D.

AI/ML • Computer Vision • Embedded Systems • ROS/ROS2

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Federal Professional Certificates: Bachelor's Degree: 12027207 • Master's Degree: 14043743 • Ph.D. Degree: 15067339

Technical Skills

AI/ML: Computer Vision, Predictive Modeling, LSTM, Time Series Analysis (ARIMA), Deep Learning (Mask R-CNN), Model Optimization
Programming Languages: Python, C++, SQL, Bash, MATLAB
Frameworks & Libraries:
C++ and Python: OpenCV, TensorFlow/Keras
Python: Scikit-learn, Pandas, Matplotlib, Seaborn, NumPy, JSON, Tkinter
Deployment & Backend: FastAPI, SQLAlchemy
Robotics Frameworks: ROS/ROS2 with C++ and Python, RViz, Gazebo
Tools: Ubuntu (Linux), VS Code, Git, GitHub, GitLab, Jupyter Notebook
Document Preparation and Office Tools: LaTeX, Markdown, MediaWiki, MS Office, Dia (technical diagram editor)

Core Strengths

- Solving complex problems in AI/ML systems, computer vision, and ROS/ROS2-based modular architectures.
- Communicating technical concepts effectively to cross-functional teams.
- Collaborating with software engineers and researchers to deliver scalable solutions.

Languages

Spanish | Native: full professional proficiency.
English | Advanced B2: professional working proficiency. Fluent in reading, writing, and technical comprehension; intermediate spoken; proficient for research publications, documentation, and international collaboration. Certified by Cinvestav, Guadalajara Campus, and Desafío Latam.

Work Experience & | Full List

Robot Programming Instructor – ROS/ROS2 — **April 2024 – Present**
Tecnológico de Monterrey (ITESM), Guadalajara — *Zapopan, Jalisco*
1) Designed and delivered hands-on courses in ROS/ROS2 using C++ and Python, focused on differential drive robots. **2)** Developed practical repositories and lecture notes covering the foundations of development, simulation, deployment, and monitoring of ROS/ROS2-based modular systems, including computer vision and autonomous navigation. **3)** Recognized as a top-rated instructor (See recognition). See repository 2025. See latest repository.
C++ · Python · ROS/ROS2 · Bash · Computer Vision · Robotics · Technical instruction

Mathematical Analyst in Information Technologies — **September 2025 – April 2026**
Geovoy, Busmen Group — *Tlajomulco de Zúñiga, Jalisco*
1) Built an automated reporting tool using LLM-powered APIs to generate analytical reports and insights. **2)** Developed statistical and machine learning models (time series analysis (ARIMA), LSTM) for route optimization and demand prediction, generating insights to improve operational decision-making. **3)** Designed and optimized APIs using FastAPI, raw SQL, and Object-Relational Mapping (ORM) to integrate statistical models into production systems for geolocation and inventory management. **4)** Deployed a MediaWiki-based platform to standardize internal processes, code, and API documentation.
Python · FastAPI · SQLAlchemy · Time Series Analysis · LSTM · Forecasting · Predictive Modeling · GPT-OSS-20B · Data Analysis · Bash

AI/Deep Learning Research Engineer – Computer Vision — **May 2021 – July 2024**
Cinvestav, Guadalajara: Medical Image Segmentation Project — *Zapopan, Jalisco*
1) Developed a deep learning model for medical image analysis, in collaboration with German research institutions, to support clinical decision-making, focusing on segmentation and quantification of retinal pathologies. **2)** Applied data preprocessing, augmentation, feature extraction, and model validation techniques to ensure robustness and reliability in healthcare-related datasets. **3)** Worked with high-resolution image datasets to improve the detection and segmentation of retinal pathologies. **4)** Contributed to a research publication in computer vision and medical imaging.
Computer Vision · Deep Learning · TensorFlow/Keras · Mask R-CNN · Image Segmentation · Data labeling, augmentation, and visualization

Computer Vision & ROS Developer — **December 2019 – October 2023**
Cinvestav, Guadalajara: Intelligent Visual Guide System (OJO SMART) — Modular Navigation Device — *Zapopan, Jalisco*
1) Developed ROS-based modules for real-time recognition of colors, objects, signs, banknotes, and text using computer vision techniques. **2)** Contributed to the integration of multiple perception systems into an assistive navigation device for visually impaired users.
Computer Vision · C++ · Python · ROS · Tesseract OCR · OpenCV · TensorFlow

Publications & Patents

Meeting Abstract | **June 2024** | "Deep-learning based quantification of RPE65-mutation inherited retinal degeneration", presented at *Investigative Ophthalmology & Visual Science*, vol. 65(7), 1392, ID: 2794864.
AI/ML lifecycle · Computer vision / Medical image analysis · Data visualization · Mask R-CNN · Feature extraction · Research

Journal Article | **September 2023** | "Quaternion and Split Quaternion Neural Networks for Low-Light Color Image Enhancement", in *IEEE Access*, vol. 11, 108257-108280, 10.1109/ACCESS.2023.3312234.
AI/ANN lifecycle · Computer vision / Image color analysis · Quaternion algebras · Color spaces · EKF

Patent | **March 2017** | "Device for controlling underactuated two-link systems with one actuator", filed under the Invention Support Program, University of Guadalajara. Application no. MX/a/2017/016436.
Embedded systems · Control theory · Digital and power electronics · PICs · SPI & I2C communication protocols

Academic Degrees

Ph.D. in Science — AI/Deep Learning | **Cinvestav, Guadalajara** — **September 2019 – May 2024**
Thesis | Deep learning for recognition and quantification of fundus pathologies using instance segmentation, and quaternion neural networks for low-light image enhancement. Download thesis.
AI/ML lifecycle · Computer vision / Image analysis · Deep Learnin · CNN/NN · Robotics · Research · Science Communication

M.Sc. — AI/Machine Learning | **Cinvestav, Guadalajara** — **September 2017 – August 2019**
Thesis | Quaternion neural networks for low-light image enhancement, and identification of an electromechanical system. Download thesis.
AI/ANNs · Computer vision · Linear algebra · Control theory · Robotics · Probability & statistics · Research · Science Communication

B.Eng. in Mechatronics — Embedded Systems | **University of Guadalajara** — **August 2012 – December 2016**
Social Service & Professional Internship | Developed embedded and mechatronic projects in an electronics and telecommunications laboratory.
Embedded systems · Linear algebra · Calculus · Control theory · Digital and power electronics · HMI · PICs & Arduino

Certifications | Additional Training

AI/DL	AI/ML	Data Cleaning	LLMs	NLP	OpenCV	Pandas
Python	R for DS	ROS	Scikit-learn	SQL	TF-Keras	Full list